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DETAIL SPECIFICATION

# PROPELLANT, GUN, HIGH ENERGY NITRAMINE

*This specification is for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all departments and agencies of the Department of Defense.*

## 1. SCOPE

**1.1 Scope.** This specification covers high energy nitramine gun propellant, Ex-99 (see 6.1).

**1.2 Classification.** The propellant is cylindrical multi-perforated grains and is a high energy nitramine propellant, similar in formulation to the Army's M43. It is designed for use in the Ex-167 charge.

## 2. APPLICABLE DOCUMENTS

**2.1 General.** The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents specified in sections 3 and 4 of this specification, whether or not they are listed.

**2.2 Government documents.**

**2.2.1 Specifications and standards.** The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

| | |
|---|---|
| MIL-DTL-398 | RDX |
| MIL-G-155 | Graphite (For use in Ammunition) |
| MIL-DTL-244 | Nitrocellulose |
| MIL-E-255 | Ethyl Centralite (Carbamite) |
| MIL-STD-1168 | Lot Numbering |
| MIL-C-005537 | Cellulose Acetate Butyrate |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Team (Code 840M), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

**AMSC N/A** **FSC 1376**

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(Copies of MIL-C-005537 are available from Commander, US Army TACOM-ARDEC, Attn: AMSTA-AR-EDE-S, Bldg 12, Picatinny Arsenal, NJ 07806-5000.)

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-129 Marking for Shipment and Storage

MIL-STD-286 Propellants, Solid; Sampling, Examination and Testing

(Unless otherwise indicated, copies of Federal and Military Specifications and Standards, are available from Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

**2.2.2 Other Government documents, drawings, and publications.** The following other government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

SPECIFICATIONS

NAVAL SEA SYSTEMS COMMAND (CAGE Code 53711)

WS 1141 Bis (2,2-Dinitropropyl) Acetal-Bis (2,2-Dinitropropyl) Formal, Mixture of

(Copies of these weapons specifications are available from Indian Head Division, Naval Surface Warfare Center, Code 8410P, 101 Strauss Ave, Indian Head MD 20640-5035.)

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199 Transportation

(The Code of Federal Regulations is available from the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402)

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (CAGE Code 53711)

10001-138439 Packing Box Mark 7 (For Smokeless Powder), General Arrangement

(Application for copies of this NAVSEA drawing should be addressed to: Commander, Naval Surface Warfare Center, Crane Division, Louisville Detachment, Attn: Code 802, Louisville, KY 40214.)

7263283 Charge, Propelling, 5-Inch Ex-167 Mod 0

(This NAVSEA drawing may be obtained from: Commander, Naval Surface Warfare Center, Indian Head Division, 101 Strauss Ave, Code 8410, Indian Head MD 20640-5035.)

NAVAL ORDNANCE STATION (CAGE Code 14083)

5755 Leverpak, Fibre Drum

(This Naval Ordnance Station drawing may be obtained from: Commander, Naval Surface Warfare Center, Indian Head Division, 101 Strauss Ave, Code 2150, Indian Head MD 20640-5035.)



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**2.3 Non-Government publications and drawings.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in solicitation. Unless otherwise specified, the issues of documents not listed in DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

| | |
|---|---|
| ASTM E 537 | Standard Test Method for Assessing the Thermal Stability of Chemicals by Methods of Differential Thermal Analysis |
| ASTM D 817 | Standard Test Methods of Testing Cellulose Acetate Propionate and Cellulose Acetate Butyrate |
| ASTM D 941 | Standard Test Method for Density and Relative Density (Specific Gravity) of Liquids by Lipkin Bicapillary Pycnometer |
| ASTM-D-1343 | Standard Test Method for Viscosity of Cellulose Derivatives by Ball-Drop Method |
| ASTM D 1744 | Standard Test Method for Determination of Water in Liquid Petroleum Products by Karl Fischer Reagent |
| ASTM D 1807 | Standard Test Method for Refractive Index and Specific Optical Dispersion of Electrical Insulating Liquids |
| ASTM-D-4795 | Standard Test Method for Nitrogen Content of Soluble Nitrocellulose - Alternative Method |
| ISO-1392 | Determination of Crystallization Point |

(Application for copies of ASTMs should be sent to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959)

**2.4 Order of Precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

**3.1 Preproduction.** Unless otherwise specified (see 6.3), a sample shall be submitted for preproduction testing (see 4.3).

**3.2 Ingredients.** The ingredients used in a specific formulation of the propellant shall meet the appropriate requirements set out in this section.

### 3.2.1 Nitrocellulose

**3.2.1.1 Nitrogen content.** The nitrogen content shall be between 12.55 and 12.65 percent with no blend component having a nitrogen content below 12.45 percent (see 4.6.1.1.1).

**3.2.1.2 Viscosity.** The viscosity shall be between 10 and 15 seconds (see 4.6.1.1.2).



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**3.2.1.3 Source.** The nitrocellulose shall be continuously nitrated and manufactured from baled cotton linters only.

**3.2.1.4 Fineness.** The fineness before dehydration shall be 80 ml ± 10 ml (see 4.6.1.1.3).

**3.2.1.5 Heat test.** The nitrocellulose shall yield a result of 35 minutes or greater both before dehydration and after packing (see 4.6.1.1.4 )

**3.2.1.6 Total volatiles.** The nitrocellulose shall be wet with 25 to 28 percent total volatiles of which a minimum of 88% shall be alcohol (see 4.6.1.1.5).

**3.2.2 RDX**

**3.2.2.1 Melting point.** The melting point of the RDX shall be no lower than 190 $^{o}$ C (see 4.6.1.2.1).

**3.2.2.2 Acidity.** The acidity of the RDX, expressed as percent acetic acid, shall be no greater than 0.02 (see 4.6.1.2.2).

**3.2.2.3 Particle size.** The average particle size shall be between 4.7 and 5.7 microns, with at least 90 percent of the particles being below 10.5 ± 0.5 microns and no more than 10 percent of the particles being less than 1.4 ± 0.1 microns (see 4.6.1.2.3).

**3.2.2.4 Purity.** No more than 0.05 percent by weight of the RDX shall consist of acetone insoluble material (see 4.6.1.2.4) and no more than 0.03 percent by weight of the RDX shall consist of inorganic insoluble material (see 4.6.1.2.4).

**3.2.3 Cellulose acetate butyrate (CAB)**

**3.2.3.1 Butyryl content.** The butyryl content of the CAB shall be between 35.0 and 39.0 percent by weight (see 4.6.1.3.1).

**3.2.3.2 Acetyl content.** The acetyl content of the CAB shall be between 12.0 and 15.0 percent by weight (see 4.6.1.3.1)

**3.2.3.3 Hydroxyl content.** The hydroxyl content of the CAB shall be between 1.2 and 2.2 percent by weight (see 4.6.1.3.1)

**3.2.3.4 Free acidity.** The free acidity content, expressed as acetic acid, shall be no more than 0.03 percent (see 4.6.1.3.2).

**3.2.3.5 Ash.** Ash shall be less than 0.05 weight percent (see 4.6.1.3.3).

**3.2.3.6 Viscosity.** The viscosity shall be between 17.0 and 24.0 seconds (see 4.6.1.3.4).

**3.2.3.7 Softening point.** The CAB shall soften between 195 and 205 $^{o}$ C (see 4.6.1.3.5).

**3.2.3.8 Moisture.** Moisture content shall be no greater than 3.0 percent by weight (see 4.6.1.3.6).

**3.2.4 Bis (2,2-dinitropropyl) acetal - bis (2,2-dinitropropyl) formal, mixture of**

**3.2.4.1 Acid number.** The acid number, expressed as milligrams of potassium hydroxide (KOH) per gram, shall be a maximum of 0.50 (see 4.6.1.5.1).

**3.2.4.2 Water content.** The weight percent of water shall be a maximum of 0.05 percent (see 4.6.1.5.2)



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**3.2.4.3 Refractive index.** The refractive index at 25°C shall be a minimum of 1.462 and a maximum of 1.464 (see 4.6.1.5.3).

**3.2.4.4 Density.** The density at 25°C shall be 1.383 to 1.397 grams per milliliter (see 4.6.1.5.4).

**3.2.4.5 Percent acetal.** The percent acetal in the mixture shall be from 45 to 55 percent (see 4.6.1.5.5).

**3.2.4.6 Percent formal.** The percent acetal in the mixture shall be from 45 to 55 percent (see 4.6.1.5.5).

**3.2.5 Ethyl centralite.**

**3.2.5.1 Solidification point.** The solidification point shall be a minimum of 71 °C and a maximum of 72.5 °C (see 4.6.1.6.1).

**3.2.5.2 Melted material.** The melted material shall be a bright liquid, free of scum (see 4.6.1.6.2).

**3.2.5.3 Volatile content.** The volatile content shall be 0.10 percent by weight or less (see 4.6.1.6.3)

**3.2.5.4 Ash content.** The ash content shall be 0.10 percent by weight or less (see 4.6.1.6.4).

**3.2.5.5 Acidity.** The acidity shall not exceed 0.04 percent by weight (see 4.6.1.6.5).

**3.2.5.6 Particle size.** 99.9 percent or more of the material shall pass through a No. 30 US sieve (see 4.6.1.6.6).

**3.2.6 Graphite.**

**3.2.6.1 Moisture.** The moisture content shall be 0.50 percent by weight or less (see 4.6.1.7.1).

**3.2.6.2 Ash.** Ash content shall be 6.0 percent by weight or less (see 4.6.1.7.2).

**3.2.6.3 Acidity.** No acidity is permitted (see 4.6.1.7.3).

**3.2.6.4 Particle size.** The particle size of the material shall be such that a minimum of 96 percent by weight passes through a No. 325 US sieve (see 4.6.1.7.4).

**3.3 Composition (M101).** The propellant shall meet the requirements of Table I. (See 4.6.2)

TABLE I. Composition

| Material | Percent by Weight |
|---|---|
| Nitrocellulose Grade A Type 1 | 4.00 ± 0.25 |
| RDX1/ | 76.00 ± 0.50 |
| Ethyl Centralite | 0.40 ± 0.04 |
| Cellulose acetate butyrate, 381-20 | 12.00 ± 0.30 |
| Bis (2,2-dinitropropyl) acetal - Bis (2,2-dinitropropyl) formal, mixture of | 7.6 ± 0.30 |
| Total volatiles | 0.20 max |
| Graphite (Applied as glazing to surface of grain) | 0.10 ± 0.05 |

1/ The RDX shall be ground in a fluid energy mill to the particle size requirements of 3.2.2.3.

**3.4 Grain form and dimensions. (M102)** The grains shall be right circular cylinders with seven cylindrical perforations parallel to the axis of the grain and evenly spaced across the circular face of the grain. The



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length of the grain shall be between 1.0 and 2.5 times the outer diameter of the grain. The length of the grain shall be no greater than fifty times the diameter of the perforation of the grain (see 4.6.3).

**3.5 Glaze.** The propellant shall be glazed with graphite, MIL-G-155, Grade IV (see 4.6.4).

**3.6 Integrity.** The propellant grains shall be without visually discernible cracks between the perforations or between the perforations and the outside edge of the grain (see 4.6.4)

**3.7 Stability (C1).** Using differential thermal analysis, the propellant shall not exhibit an exotherm onset at a temperature lower than $210^{o}$ C. (See 4.6.5). (Check this against DTA test results.)

**3.8 Packing depth (M103).** The production packing depth of the assessed charge weight shall be between 5 and 10 centimeters measured from the case mouth (see 4.6.6).

**3.9 Gun ballistic assessment requirements**

**3.9.1 Velocity (M104).** The charge weight of the propellant lot under test shall be assessed to result in a nominal average new gun velocity of 850 m/s. (See 4.6.7)

**3.9.2 Pressure (M105)** The average peak chamber pressure shall be no more than 448 MPa (piezoelectric pressure measurement). (See 4.6.7)

**3.9.3 Velocity variation (M106)** The maximum allowed velocity standard deviation shall be 8.5 m/s. (See 4.6.7).

**3.9.4 Pressure variation (M107)** The maximum allowable peak pressure standard deviation shall be 14 MPa. (See 4.6.7)

**3.10 Propellant interface test.** All preproduction lots and production lots, when directed by the procuring activity, must be subjected to the propellant interface test (see 4.6.10 and 6.4)

## 4. QUALITY ASSURANCE PROVISIONS

**4.1 Classification of inspections.** The inspection requirements specified herein are classified as follows:

a. Preproduction inspection (See 4.3).
b. Quality conformance inspection (See 4.4).

**4.2 Inspection conditions.** All test activities identified in the contract shall submit to the technical activity and the design agent for review, prior to first article (preproduction), a description of the proposed procedures and equipment to be used to perform the examination and test.

**4.3 Preproduction inspection.** Preproduction inspection (except for production packing depth determination and ballistic testing) shall be performed by the contractor, after award of contract and prior to production, at a location acceptable to the Government. Preproduction inspection shall be performed on sample material which has been produced with equipment and procedures normally used in production. Preproduction approval is valid only on the contract under which it is granted, unless extended by the Government to other contracts. The preproduction sample is intended to provide necessary information to establish the propellant granulation needed to meet the required ballistic parameters. A preproduction sample should be required any time a change is made to the formulation of the propellant, one of its ingredients, the production process for the propellant or certain changes to the propelling charge design expected to alter the ballistics.



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**4.3.1 Submission.** The Contractor shall select three propellant granulations which, from experience and/or analysis, are expected to bracket the ballistic requirements of this specification. The contractor shall submit a preproduction sample as designated by the Contracting Officer for evaluation in accordance with 4.4.2, consisting of three pilot lots of between 350 and 450 kg each, representing the three candidate granulations.

**4.3.2 Inspections to be performed.** Each of the granulations comprising the preproduction sample and submitted in accordance with 4.3.1 shall be tested by the Government. Preproduction inspection shall consist of the tests specified in Table II.

TABLE II. Preproduction and quality conformance tests.

| Characteristic | Requirement | Test Method |
|---|---|---|
| Composition | 3.2 | 4.6.1 |
| Grain form and dimensions | 3.4 | 4.6.2, 4.6.3 |
| Glaze | 3.5 | 4.6.3 |
| Integrity | 3.6 | 4.6.3 |
| Stability | 3.7 | 4.6.4 |
| Packing depth | 3.8 | 4.6.5 |
| Velocity and pressure | 3.9 | 4.6.6 |

**4.3.3 Authorization.** Government Engineering and Product Assurance personnel and the Contractor shall review the pilot lot test data and shall select a granulation size which best fits the requirements of this specification. The Technical Agency shall notify the Contracting Officer of the selected granulation. The Contractor shall request approval from the Contracting Officer to produce the selected granulation. If the pilot lots submitted to meet the requirements of this specification do not bracket the required performance, a new preproduction sample shall be submitted in accordance with 4.3.1. All future production lots under this contract line item shall be of the same basic granulation.

**4.3.4 Failure.** Failure of the sample to meet the requirements of this specification shall be cause for rejection of the preproduction sample.

**4.4 Quality conformance inspection.** Quality conformance inspection shall consist of the examinations and tests of 4.6 to determine conformance to the requirements of this specification. Failure of any sample to conform to any requirement specified herein shall be cause for rejection of the lot represented. Alternate methods and test procedures may be used provided prior written approval has been obtained from the procuring activity and the gun ammunition propelling charge design agent, Indian Head Division, Code 621, 101 Strauss Avenue, Naval Surface Warfare Center, Indian Head, Maryland. Requirements and their corresponding tests are summarized in Table II.

**4.4.1 Lot size.** Unless otherwise specified in the contract or purchase order, the propellant shall be produced in lots of not less than 10,000 nor greater than 250,000 kilograms (kg).

**4.4.2 Sampling.** The samples shall be selected from full Mark 7 boxes (Drawing 138439) or leverpaks (Drawing NOS 5755) so as to be representative of the lot. All samples shall be selected following the final packing of the lot. All samples shall be packed in moisture-proof airtight containers and marked with the following in accordance with MIL-STD-129:

a. Propellant description
b. Propellant lot number
c. Number of kilograms in lot
d. Manufacturer
e. Contract number
f. Net weight in container



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**4.4.2.1 For gun performance tests.** Sample size shall be approximately 1100 kg randomly selected from full Mark 7 boxes or leverpaks.

**4.4.2.2 For physical characteristic tests.** A 5-kg sample shall be randomly selected from full Mark 7 boxes or leverpaks.

**4.4.2.3 For surveillance tests.** A 3-kg sample shall be randomly selected from full Mark 7 boxes or leverpaks and forwarded to the Navy's designated surveillance activity for propellants; Indian Head Division, Naval Surface Warfare Center, Attn: Code 621F, 101 Strauss Avenue, Indian Head, MD 20640-5035.

**4.5 Classification of characteristics.** The characteristics verified by the tests and examinations herein are classified as critical, major, or minor in accordance with DOD-STD-2101. Requirements for critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of the characteristic. Requirements which are not annotated with a classification code are classified minor.

**4.6 METHODS OF INSPECTION.**

PRECAUTION - This specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel.

**4.6.1 Ingredient analyses.** The ingredients shall be analyzed as described in this section. The requirements of 3.2 shall be met or the ingredients shall not be used in making the propellant. Propellant made with ingredients which do not meet the requirements of 3.2 shall be rejected.

**4.6.1.1 Nitrocellulose inspections.**

**4.6.1.1.1 Nitrogen content.** The nitrogen content shall be determined using ASTM-D-4795. The requirements of 3.2.1.1 shall be met or the material shall be rejected for use in making the propellant.

**4.6.1.1.2 Viscosity.** The viscosity shall be determined using ASTM-D-1343. The requirements of 3.2.1.2 shall be met or the material shall be rejected for use in making the propellant.

**4.6.1.1.3 Fineness.** The fineness before dehydration shall be determined by taking a 10-ml specimen, previously dried at 100° to 105° C, and weighed to the nearest decigram, and placing that specimen in a 250-ml beaker. A 100 to 150-ml portion of distilled water shall be added and mixed gently rotating the beaker. The mixture shall be transferred to a 250-ml glass-stoppered measuring cylinder, graduated in 2-ml divisions over a length of approximately 25 cm. The beaker shall be rinsed several times with small washes of distilled water, adding each wash to that present in the cylinder until the suspension measures approximately 200 ml. The cylinder shall be stoppered and shaken vigorously with a vertical motion. The stopper shall be removed and the stopper and sides of the cylinder shall be rinsed down until the volume of the suspension reaches 250 ml. The stopper shall be replaced and the cylinder allowed to stand for one hour. At the end of this time, the volume in ml occupied by the nitrocellulose, shall be read and this value shall be considered as the fineness. The requirements of 3.2.1.4 shall be met or the material shall be rejected for use in making the propellant.

**4.6.1.1.4 Heat test.** The stability of the nitrocellulose under mild heating shall be tested in accordance with MIL-N-244 section 4.4.8. The requirements of 3.2.1.5 shall be met or the material shall be rejected for use in making the propellant.

**4.6.1.1.5 Total volatiles.** The total volatiles in the nitrocellulose shall be determined in accordance with MIL-N-244 section 4.4.9. The requirements of 3.2.1.6 shall be met or the material shall be rejected for use in making the propellant.



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**4.6.1.2 RDX Inspections**

**4.6.1.2.1 Melting point.** The melting point of the RDX shall be determined in accordance with ISO-1392. The requirements of 3.2.2.1 shall be met or the material shall be rejected for use in making the propellant.

**4.6.1.2.2 Acidity.** The acidity of the RDX shall be determined in accordance with MIL-R-398 section 4.3.5. The requirements of 3.2.2.2 shall be met or the material shall be rejected for use in making the propellant.

**4.6.1.2.3 Particle size.** A Microtrac analyzer shall be used for determining particle size. The requirements of 3.2.2.3 shall be met or the material shall be rejected for use in making the propellant.

**4.6.1.2.4 Purity.** The amount of material insoluble in acetone and inorganic insolubles shall be determined in accordance with MIL-R-398 sections 4.3.2 and 4.3.3 respectively. The requirements of 3.2.2.4 shall be met or the material shall be rejected for use in making the propellant.

**4.6.1.3 Cellulose acetate butyrate inspections**

**4.6.1.3.1 Butyryl, acetyl and hydroxyl content .** Determine the butyryl, acetyl and hydroxyl contents using the method given in ASTM D 817 sections 13 through 39. The requirements of 3.2.3.1, 3.2.3.2 and 3.2.3.3 shall be met or the material shall be rejected for use in making the propellant.

**4.6.1.3.2 Free acidity.** Determine free acidity as directed in ASTM D 817 sections 8 through 12. The requirements of 3.2.3.4 shall be met or the material shall be rejected for use in making the propellant.

**4.6.1.3.3 Ash.** Determine ash content as directed in ASTM D 817 sections 6 and 7. The requirements of 3.2.3.5 shall be met or the material shall be rejected for use in making the propellant.

**4.6.1.3.4 Viscosity.** Viscosity shall be determined by ASTM D 1343. The requirements of 3.2.3.6 shall be met or the material shall be rejected for use in making the propellant.

**4.6.1.3.5 Softening point.** The softening point shall be determined with a hot stage melting point apparatus. The requirements of 3.2.3.7 shall be met or the material shall be rejected for use in making the propellant.

**4.6.1.3.6 Moisture.** Determine the moisture content as directed in ASTM D 817 sections 4 and 5. The requirements of 3.2.3.8 shall be met or the material shall be rejected for use in making the propellant.

**4.6.1.4 Bis (2,2-dinitropropyl) acetal-bis (2,2-dinitropropyl) formal (BDNPA/F)(mixture of) inspections**

**4.6.1.5.1 Acid number.** The acid number shall be determined as in a. through f. The requirements of 3.2.4.1 shall be met or the material shall be rejected for use in making the propellant.

a. Draw approximately 20 grams of BDNPA/F and weight it to the nearest 0.005 grams into a 150-milliliter Erlenmeyer flask.

b. To the flask add 50 milliliters of a 1:1 solution of methanol/chloroform, 50 milliliters of distilled water and 10 drops of phenolphthalein solution.

c. Place magnetic stir bar in flask, place on magnetic stirrer and agitate for two minutes.

d. Titrate with 0.02N KOH to a 30-second pink end point in the water layer.

e. Run blank with all reagents omitting BDNPA/F.



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f. Calculate acid number as follows:

Acid Number (mg KOH/g) = [(S-B) (N) (56.11)]/W

Where S = Volume of KOH for sample
B = Volume of KOH for blank
N = Normality of KOH
W = Weight of sample

**4.6.1.4.2 Water content.** Water content shall be determined as described in MIL-STD-286 method 101.5. The requirements of 3.2.4.2 shall be met or the material shall be rejected for use in making the propellant.

**4.6.1.43 Refractive index.** The refractive index shall be determined according to the method described in ASTM D 1807. The requirements of 3.2.4.3 shall be met or the material shall be rejected for use in making the propellant.

**4.6.1.4.4 Density.** Density shall be determined according to the method described in ASTM D 941. The requirements of 3.2.4.5 shall be met or the material shall be rejected for use in making the propellant.

**4.6.1.4.5 Acetal and formal contents.** The acetal and formal contents shall be determined according to the method described in a. through h. The requirements of 3.2.4.5 and 3.2.4.6 shall be met or the material shall be rejected for use in making the propellant.

a. This procedure requires a high performance liquid chromatograph (HPLC). The following solvents and equipment are suggested but are not required. Any comparable solvents or equipment may be substituted.

b. The suggested equipment includes a Waters 990 photodiode array detector, Waters 600E system controller, Waters intelligent sample processor (WISP 712), Waters radical pak 5 micron C18 column, and a Waters 8x10 compression module. All solvents are HPLC grade, manufactured by Burdick and Jackson Laboratories. The water is obtained from a Milli-Q laboratory water system.

c. The following HPLC parameters may be used:

| | |
|---|---|
| Wavelength range | 190-33 nanometers |
| Monitor wavelength | 280 nanometers |
| Sensitivity | -0.03 to 1.0 aufs |
| Interval | 0.2 sec |
| Resolution | 3 nanometers |
| Sense | Normal |
| Smoothing | 7 points |
| Slope | 0.01 au/min |
| Drift | 0.01 au/min |
| Height | 0.0001 au/min |
| Width | 0.01 min |
| Minimum area | 0.001 au/min |

d. The suggested mobile phase for the determination of the acetal/formal content is 40:40:20, methanol:water:2-propanol, at a flow rate of 3.0 ml/min with a 50 milliliter helium sparge. A run time of 6 minutes may be used with specified conditions and radial pak column. The calibration standard is prepared by weighing 0.25 grams each of the BDNPA and BDNPF, weighed to the nearest 0.1 milligram, into a 10 milliliter volumetric flask and diluting to volume with methanol.



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e. Prepare a calibration table by averaging the integrated areas of three 40-microliter injections of the standard solution. Prepare this sample solution by weighing 0.50 grams, weighed to the nearest 0.1 milligram, of the BDNPA/F into a 10-milliliter volumetric flask and diluting with HPLC-grade methanol.

f. Make three 40-microliter injections of the sample solution, comparing the area of the sample to the areas of the standard BDNPA and BDNPF.

Calculate the percent acetal or formal as follows:

A/S (100) = percent acetal or formal

Where A = milligrams of acetal or formal as integrated
S = sample weight of the BDNPA/F

Note: The HPLC conditions may require alterations to achieve the desired resolution. This largely depends on the individual column.

**4.6.1.5.5.6 BDNPA Standard.** Preparation of BDNPA reference standard shall be as described in a. through e.

a. The reference standard shall be prepared by placing 450.3 g (3.0 moles) of 2,2-dinitropropanol (DNPOH), 105.5 g (2.4 moles) of acetaldehyde, and 1470 g of ethylene chloride into a three-necked flask equipped with a cooling bath, thermometer and dropping funnel. Stir the solution continuously, keeping the temperature between -11$^{o}$C and -9$^{o}$C. Add 213 g (1.5 moles) of boron trifluoride etherate over a 15-minute period of time. Add 15 to 30 g initially and cool until the exothermic reaction is under control; then add the remainder. Stir the mixture for 15 minutes at 10$^{o}$C and then quench with 930 g of water.

b. Allow the two-phase mixture to settle after 10 minutes of agitation. Separate the aqueous layer and discard. Wash the organic layer with sufficient 4 percent sodium hydroxide solution to raise the pH to 11.0. Control the pH at 11.0 by the intermittent addition of sodium hydroxide over a period of 20 to 30 minutes stirring time. Separate the layers and wash the organic layer again with an equal volume of 5 percent sodium hydroxide solution at a temperature of 10 to 15$^{o}$C for 30 minutes. Discard the aqueous layer and wash the organic layer with one-liter portions of water until neutral.

c. Remove the solvent in vacuum at 45$^{o}$C and 0.1 mm mercury. Dissolve the crude BDNPA in 0.84 g of isopropyl ether per 1.0 g of BDNPA and crystallize at -30$^{o}$C for 2 hours. Filter the crystals by suction and dry at 45$^{o}$C and 0.1 mm mercury for 2 hours.

d. Charge the crystallized BDNPA into a 2-inch Arthur F. Smith high vacuum still and distill twice at 125$^{o}$C and 10 microns mercury.

e. Set up the 2-inch Arthur F. Smith high vacuum wiped-film still as follows in e.1 through e.5.

1 Set up the regular vapor cold trap and diffusion pump supplied with the unit as directed. Attach the manometer between the still and the vapor trap. Attach a vacuum pump protected by two cold traps in series to the diffusion pump. Secure a 500-ml round bottom flask receiver constructed with a bottom and overhead outlet to the residue outlet at the bottom of the still. Connect the bottom outlet of this flask to the top outlet of a similar flask just below with a 1/4-inch latex vacuum hose equipped with a pinch clamp. By means of rubber tubing, use the top outlet of the residue receiver to connect a pressure equalizing line to the neck of the lower feed container.

2 Use a sigmamotor pump to transfer the BDNPA from the bottom outlet of the feed container through a glass condenser heat exchanger to the feed inlet of the wiped-film still through



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1/4-inch latex vacuum tubing. Cool both the residue received and the feed container by circulating cold water in and out of the cooling pans supporting these flasks.

3 Attach a suitable distillate receiver to the distillate outlet of the still. Circulate the fluid through the condenser finger in the still and heat exchanger by means of an immersion pump.

4 Wrap a glascol heating mantle around the still and measure the skin temperature between the still and the mantle by means of a thermocouple temperature indicator. For the most effective start-up procedure, check all connections for leaks prior to charging the feed container with the BDNPA.

5 Charge the cold traps with dry ice and activate the vacuum pump. Turn on the diffusion pump when the pressure is 1 to 2 mm mercury. Adjust the sigmamotor pump to 1 to 1.5 liters per hour and activate the still's wiping blades. Adjust the still heating mantle to 120 °C to 125 °C skin temperature and the condenser and heat exchanger fluid to 40 °C to 50 °C. The first pass through the still is usually sufficient to degas the BDNPA. When the feed container is empty, release the pinch clamp and let the degassed material from the receiver enter the feed container. After the BDNPA is completely degassed and distilling properly, the pinch clamp can be removed; but if left in use, the number of passes through the still to complete the distillation can be determined.

**4.6.1.4.5.7 BDNPF Standard.** Preparation of BDNPF reference standard shall be as described in a. through e.

a. The reference standard shall be prepared by adding 225 g (2.25 moles) of 98 percent sulfuric acid over a 15-minute period to a continuously stirred solution of 473.0 g (3.15 moles) of 2,2-dinitropropanol (DNPOH) and 45 g (0.5 moles) of trioxane in 1470 g of ethylene chloride. Stir for 45 minutes at 25 °C.

b. Allow the mixture to settle and separate the spent sulfuric acid from the organic layer. Wash the organic layer twice with one half its volume of five percent sodium hydroxide. During the two washings, dwell for 30 minutes at 15 °C and 25 °C respectively. Separate the aqueous layer and discard. Wash the organic layer with one-liter portions of water until neutral.

c. Strip the solvent in vacuum at 45 °C and 0.1 mm mercury for two hours. Dissolve the crude BDNPF in 1.4 g of denatured alcohol (a mixture of 100 parts by volume no. 1 denatured ethanol, 10 parts isopropanol and one part methyl isobutyl ketone) per 1.0 g of BDNPF and crystallize at -30 °C for 2 hours. Filter the crystals by suction and dry at 45 °C and 0.1 mm mercury for 2 hours.

d. Charge the crystallized BDNPF into a 2-inch Arthur F. Smith high vacuum still and distill twice at 125 °C and 10 microns mercury. The reference standard need not be prepared fresh for each determination.

e. Set up and operate the 2-inch Arthur F. Smith high vacuum wiped-film still in accordance with 4.6.1.4.5.6.e.1 through 4.6.1.4.5.6.e.5.

**4.6.1.5 Ethyl centralite inspections**

**4.6.1.5.1 Solidification point.** The solidification point shall be determined in accordance with the method described in MIL-E-255 paragraph 4.3.1. The requirements of 3.2.5.1 shall be met or the material shall be rejected for use in making the propellant.

**4.6.1.5.2 Melted material.** The appearance of the melted material shall be determined visibly during the test of 4.6.1.5.1. The requirements of 3.2.5.2 shall be met or the material shall be rejected for use in making the propellant.



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**4.6.1.5.3 Volatile matter.** A weighed portion of approximately 5 grams of the sample shall be transferred to a tared shallow dish, heated for two hours at 60°C, cooled in a desicator and weighed. The loss in weight shall be calculated as percentage of volatile matter in the sample. The requirements of 3.2.5.3 shall be met or the material shall be rejected for use in making the propellant.

**4.6.1.5.4 Ash.** The ash content shall be determined as directed in MIL-STD-286 method 106.1.2. The requirements of 3.2.5.4 shall be met or the material shall be rejected for use in making the propellant.

**4.6.1.5.5 Acidity.** The acidity shall be determined in accordance with the method described in MIL-E-255 paragraph 4.3.6. The requirements of 3.2.5.5 shall be met or the material shall be rejected for use in making the propellant.

**4.6.1.5.6 Particle size.** The particle size shall be determined in accordance with the method described in MIL-E-255 paragraph 4.3.8.1. The requirements of 3.2.5.6 shall be met or the material shall be rejected for use in making the propellant.

**4.6.1.6 Graphite inspections**

**4.6.1.6.1 Moisture.** Moisture shall be determined in accordance with MIL-STD-286 method 101.2.2. The requirements of 3.2.6.1 shall be met or the material shall be rejected for use in making the propellant.

**4.6.1.6.2 Ash.** The ash content shall be determined as directed in MIL-STD-286 method 106.1.2. The requirements of 3.2.6.2 shall be met or the material shall be rejected for use in making the propellant.

**4.6.1.6.3 Acidity.** Acidity shall be determined by transferring a 10-gram sample of material to a 250-ml beaker, adding 100 ml of neutral distilled water and heating quickly to boiling while stirring. Filter immediately and cool to room temperature. Test the filtrate by adding two drops of phenolphthalein and then two drops of methyl red. If colorless to phenolphthalein and yellow to methyl red, acidity shall be reported as none. The requirements of 3.2.6.3 shall be met or the material shall be rejected for use in making the propellant.

**4.6.1.6.4 Particle size.** The particle size shall be determined in accordance with MIL-G-155 paragraph 4.3.8. The requirements of 3.2.6.4 shall be met or the material shall be rejected for use in making the propellant.

**4.6.2 Chemical Composition.** The chemical composition of the propellant shall be witnessed by certified production mix documentation. The requirements of 3.3 shall be met or the lot shall be rejected. A chemical analysis performed according to the methods cited in Table III shall be performed for information only.

TABLE III. Chemical analysis test methods.

| Ingredients | Test Methods |
|---|---|
| RDX 5.5 μ FEM | MIL-P-70818 para 4.5.1.1 |
| Cellulose acetate butyrate | MIL-P-70818 para 4.5.1.14 |
| Nitrocellulose | MIL-P-70818 para 4.5.1.14 |
| Bis (2,2-dinitropropyl) acetal- bis (2,2-dinitropropyl) formal, mixture of | MIL-P-70818 para 4.5.1.1 |
| Ethyl Centralite | MIL-P-70818 para 4.5.1.1 |
| Total volatiles | MIL-STD-286C Method T103.5 |
| Graphite | MIL-STD-286C Method 308.1.4 |

**4.6.3 Grain Dimensions.** Grain dimensions shall be determined in accordance with MIL-STD-286, method 504.1.1 or 504.6.1. The requirements of 3.4 shall be met or the lot shall be rejected.



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**4.6.4 Visual Examination.** The propellant grains selected for the physical characteristic tests shall be visually examined prior to conducting the physical characteristic tests (4.6.3 and 4.6.5) to verify that:

a. The grain form conforms to the requirements of 3.4 and 3.6.
b. The grains are uniformly coated with graphite, conforming to the requirements of 3.5.

**4.6.5 Stability**. Stability shall be determined in accordance with ASTM E 537. The requirements of 3.7 shall be met or the lot shall be rejected.

**4.6.6 Packing Depth.** The production packing depth (PPD) is the distance, measured to the nearest 3 mm, from the case mouth to the upper level of any given propellant charge in a loaded cartridge case. A chart of loading depth versus charge weight will be determined at the load, assemble and pack (LAP) facility by loading the lot on the same equipment as will be used to load the production charges. PPD is determined by pouring the propellant charge in the same manner as it will be loaded, i.e., from a height of approximately one and a half meters above the case mouth, through a tube only slightly larger than the case. The PPD must be determined for each lot of propellant by pouring three charges (the propellant for which has been preconditioned for a minimum of four days at 21°C $\pm$ 2°C ) at each of three charge levels covering the range expected to be used in acceptance tests of the lot of propellant. A curve shall be fitted through the means of the data. The PPD associated with the charge weight assessed by the proving ground shall meet the requirements of 3.8 or the lot will be rejected.

**4.6.7 Ballistic Testing**

**4.6.7.1 Warming or Conditioning Rounds.** Warming or conditioning rounds shall be fired to stabilize gun temperature and to eliminate the biasing effects of firing the propellant at various charge levels or temperatures. The first round fired in a gun during a test is the warming round. The first round fired as part of a temperature conditioning group or charge level group wherein the velocity is expected to differ from the previous group by approximately 30 m/s or more is a conditioning round. Warming and conditioning rounds may have unusual ballistics and therefore the results are disregarded. These rounds are in addition to those specified in subsequent paragraphs.

**4.6.7.2 Gun Performance Tests.** All ballistic testing shall be conducted using fleet-released 5-inch, 62-caliber gun barrels which have met the acceptance criteria during proofing and which have a velocity loss of 15 m/s or less. The Mark 45 Mod 4 gun system shall be used except for the charge weight probe and validity tests. The non-test components (e.g.; case, primer, etc.) of the complete round of ammunition shall be from one acceptable lot per component and the cartridge shall be assembled in accordance with the applicable Naval Sea Systems Command Propelling Charge Assembly drawing, 7263283 Charge, Propelling, 5-Inch Ex-167 Mod 0 (see 2.1). Deviations in propelling charge assembly shall consist of adjustment of charge weight/production packing depth (PPD) to vary ballistics. The projectiles shall be inert loaded and their weight shall be 49.9 + 0.1 kg. The chamber volume resulting from the projectile and propelling charge combination shall be representative of that of the fleet use projectile/propelling charge combination. The ballistic test shall be performed on the preproduction and production lot except that:

a. For the preproduction sample, the test of 6.4 herein is mandatory.

b. For the production lot, the test of 6.4 herein shall be performed at the discretion of the government when the government deems the test necessary.

**4.6.7.3 Charge Assessment.** Charge assessment shall be accomplished by conducting the firing programs described below. Only one gun shall be fired each day and the velocities shall be corrected to the muzzle. The cases, primers and propellant for all propelling charges shall be preconditioned at 21°C $\pm$ 2°C for a minimum of fours days immediately prior to assembly and shall be fired at 21°C $\pm$ 2°C. Charges shall be loaded to yield the packing depth within 1 cm of that determined in 4.6.6 for the charge weight to be fired. The charges should be crimped, though the use of adhesive is not necessary. For the charge weight validity firing program the cartridges shall be reconditioned after assembly for a minimum of 48



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hours at 21°C ± 2°C prior to firing. For all tests, the cartridges shall be fired within five (5) minutes of removal from temperature conditioning.

**4.6.7.3.1 Charge Weight Probe.** A probing round firing program shall be conducted in an instrumented test mount to determine the estimated charge weight of the test propellant (TP) needed to match the required velocity. The test shall consist of firing a series of TP rounds wherein the TP charge is adjusted to vary the velocity. Sufficient rounds should be fired to obtain at least two rounds below and two rounds at or above the required velocity. An estimate shall be made from the TP velocity-charge data of the TP charge weight required to match the required velocity (C1). If the estimated pressure for the estimated TP charge, as determined from the pressure-charge data, exceeds the maximum service pressure limit of 3.9.2 by more than 11 MPa, the probing round program shall be repeated. If the estimated TP charge again has an estimated pressure which exceeds the maximum chamber pressure by 11 MPa, the lot of propellant shall be rejected without further testing.

**4.6.7.3.2 Powder Proof Firings.** A firing program shall be conducted in each of four gun barrels in the automatic Mk 45 gun system to determine the charge of the TP required to match the required velocity. Three TP charges at each of four charge levels shall be fired in each gun barrel. The first TP charge ($C_1$) shall be that which was determined during the probing round test to match the required velocity. The second charge ($C_2$) shall be approximately 0.25 kg less than $C_1$ and third charge ($C_3$) shall be approximately 0.25 kg greater than $C_1$. The fourth charge or matching charge (MC) is that charge of the TP which is predicted to match the required velocity on that day in that gun.

**4.6.7.3.3 Data Reduction.** A linear, least squares equation shall be calculated for the velocity-charge data for each gun. The MC is determined by substituting the required velocity in the equation and solving for the charge weight.

**4.6.7.3.4 Assessed Charges.** The assessed charge (AC) shall be determined by using the formula:

$$AC = \frac{\sum_{i=1}^{4} (MC_i/\Delta_i^2)}{\sum_{i=1}^{4} (1/\Delta_i)^2}$$

where: MC = charge of the test propellant estimated to match the required velocity in the $i^{th}$ gun.

$\Delta_i$ = difference between the mean muzzle velocity obtained with the matching charge in the $i^{th}$ gun and the required velocity. (If $\Delta_i$ is zero, replace the zero with 1.) The velocity shall meet the requirements of 3.9.1

**4.6.7.3.5 Charge Weight Validity.** The validity of the assessed charge shall be determined by a firing program in any of the gun barrels used in the powder proof firings from the instrumented test mount with cases instrumented with piezoelectric gauges for breech pressure measurement. The program shall consist of 10 TP rounds which have been assembled and preconditioned as noted 4.6.7.3 except that adhesive shall be used and cured. Velocity measured in this test is for information only. This mean pressure determined in this program is considered to be the mean pressure of the charge. The mean pressure and pressure standard deviations shall meet the requirements of 3.9.2 and 3.9.4 respectively or the lot shall be rejected.

**4.6.7.4 Velocity uniformity.** The velocity standard deviation shall be determined from the matching charge data from each gun barrel. The velocity standard deviation shall meet the requirements of 3.9.3 or the lot shall be rejected. To calculate the velocity standard deviation, perform the calculations described in a through d.

a. Calculate the sum of the squares of the deviations from the mean for each gun.



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b. Sum across the guns.

c. Divide by the quantity of the sum of the total number of rounds minus the number of guns.

d. Take square root to nearest unit of acceptance criteria.

**4.6.8 Examination of Preparation for Delivery.** Each unit of packaging shall be visually examined to ensure that containers are properly closed, labeled and free of major dents or breaks.

**4.6.9 Relative quickness.** Relative quickness of the propellant shall be conducted in a closed bomb as specified in MIL-STD-286, Method 801.1.2. The reference propellant shall be representative of the nominal composition of the propellant type to be tested. The "hot-wire" method of ignition may be used as an alternative to the "squib" method. This test is done for information only.

**4.6.10 Interface tests.** The procedure is described in the notes (see 6.4). For the preproduction lot the test is mandatory. For the production lot, the test shall be performed at the discretion of the Government.

## 5. PACKAGING

**5.1 Packaging.** For acquisition purposes, the packaging requirements shall be specified in the contract or order (see 6.2, 6.7). When actual packaging of material is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military's Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

**5.2 Lot Numbering.** Lot numbering shall be in accordance with MIL-STD-1168.

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

**6.1 Intended Use.** The propellant covered by this specification is intended for use in the 5-inch, 62-caliber gun system. The charge is the Ex-167 for firing the Ex-171 Extended Range Guided Munition. This propellant is military unique because it is used in Navy ship's gun ammunition only and has no commercial application. This is a detail specification because it deals with an explosive material for which safety considerations are paramount. Development and qualification of the explosive required years of testing and evaluation to ensure that the material's sensitivity to handling stimuli is not overly high, that the material exhibits adequate stability and sufficient safe life and that it is chemically compatible with the other components of the propelling charge with which it may come in contact. Deviations from this specification, especially in the ingredients and composition, could incur severe explosive hazards.

**6.2 Approved POP Packages**. The currently approved POP packages for this material are the Mk 7 Powder Can, Drawing 138439 and the leverpak, Drawing 5755.

**6.3 Ordering Data.** Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2)



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c. If preproduction is required and, if so, sample size (See 3.1 and 4.4.1)

d. Lot size if other than 4.5.1

e. Activity to conduct gun performance tests, generally Dahlgren Division, Naval Surface Warfare Center (see 4.7.7.2)

f. Packaging and marking considerations (see 5.1, 6.7)

g. Data required (see 6.4)

h. Activity to receive surveillance sample, generally Naval Surface Warfare Center, Indian Head Division (see 4.5.2.3)

**6.3 Description Sheets and Test Reports**

**6.3.1 Propellant Description Sheets.** The propellant manufacturer should furnish a propellant description sheet in accordance with MIL-STD-1171 giving a complete history of the manufacture of each propellant lot and the results of all tests performed on it. Unless otherwise specified, copies of each description sheet should be submitted to each of the following: The Navy's gun ammunition propelling charge design agent, Indian Head Division, Naval Surface Warfare Center (Code 621C), 101 Strauss Ave, Indian Head, MD 20640-5035; Crane Division, Naval Surface Warfare Center (Code 4025), Crane, IN 47522-5050; and to the activity designated to perform the gun performance tests (generally Dahlgren Division, Naval Surface Warfare Center, Code G61).

**6.3.2 Gun Performance Test Reports.** The activity which conducts the gun performance tests (4.7.6.2) should furnish a complete test report, including all test, gun, projectile, and cartridge components data. Unless otherwise specified, two copies of each test report should be submitted to the following: Crane Division, NSWC, Code 4025, Crane, IN 47522-5050; Indian Head Division, NSWC, Code 621, 101 Strauss Avenue, Indian Head, MD 20640-5053.

**6.4 Propellant Interface Test.** A test firing program should be conducted in an instrumented mount (instrumentation location to be as specified in NSWC, Dahlgren Laboratory - TOE Drawing 1002) on the preproduction lot to verify that there is no adverse primer-propellant relationship which could cause problems in the fleet. The test would consist of twelve data rounds utilizing the assessed charge weight from 4.6.7.8. Three rounds would be preconditioned for a minimum of 72 hours and fired at each of three temperatures: -7° $\pm$ 2° C, 10° $\pm$ 2° C, and 49° $\pm$ 2°C. In addition, a minimum of one data round per temperature of the applicable master propellant would be fired for comparison. The non-test components (e.g., case, primer, etc.) of the propelling charge and the propelled mass of 50.0 $\pm$ 0.1 kilograms would be from the same acceptable lots used for the ballistic tests (see 4.6.7.3). All rounds would be assembled in accordance with the applicable Naval Surface Warfare Center, Indian Head Division Propelling Charge Assembly drawing (see 2.1) except for the required instrumentation gauges. The analyses of the pressure-time plots would be performed by the Navy's gun ammunition propelling charge design agent, Code 621C, Indian Head Division, NSWC, Indian Head, Maryland. Subsequently, upon completion of analyses, the design agent would advise the Naval Surface Warfare Center, Crane Division in writing, of the results of these analyses.

**6.5 Advisory.**

**6.5.1 Relative quickness.** The average relative quickness value should be within 3.0 percent of the reference lot when measured in a closed bomb in order to be certain of meeting the required ballistics (see 4.7.9)

**6.5.2 Heat of Explosion (HOE).** Experience indicates that properly performing propellant demonstrates a nominal heat of explosion of 955 cal/g.



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**6.5.3 Density.** Experience indicates that properly performing propellant demonstrates a density not less than 1.65 g/cc.)

**6.6 Safety Precautions.** Safety precaution requirements of DOD 4145.26 M are applicable. Note: When this specification is used as part of work to be accomplished by a Government activity, the safety precaution requirements of OP 5 are applicable. For Army activities, the safety precautions of AMCR-385-100 are applicable.

**6.7 Packaging**

**6.7.1 Packing.** Packing should be level A. The propellant should be packed in accordance with the packing requirements of 49 CFR Part 171-178, which states that containers must be approved according to United Nations Performance Oriented Packaging (POP). (See 6.2)

**6.7.2 Marking.** In addition to any special marking required by the contract or order, shipping containers should be marked in accordance with the requirements of 49 CFR Part 171-178. Proper shipping name and serial number in accordance with CFR 49 is "PROPELLANT, SOLID, UN 0499". The appropriate United Nations Performance Oriented Packaging marking should be applied on side of container opposite the identification marking with letters .50 inches minimum height. Encirclement of U/N symbol should be sufficiently large to provide a minimum clear spacing around letters. (**) insert year packed.

**6.8 Subject term (key word) listing.**

Propelling Charge
5-inch
62 Caliber
Extended Range
Guided Munition

Preparing Activity:
Navy – OS
(Project 1376-0071)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER<br>MIL-DTL-82965A | 2. DOCUMENT DATE (YYMMDD)<br>991220 |
|---|---|---|

3. DOCUMENT TITLE
PROPELLANT, GUN, HIGH ENERGY NITRAMINE

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (*Last, First, Middle Initial*) | b. ORGANIZATION | |
|---|---|---|
| C. ADDRESS (*Include Zip Code*) | d. TELEPHONE (*Include Area Code*)<br>(1) Commercial<br><br>(2) DSN<br>(*If applicable*) | 7. DATE SUBMITTED<br>(*YYMMDD*) |

8. PREPARING ACTIVITY

| a. NAME<br>Commander, Indian Head Division<br>Naval Surface Warfare Center | b. TELEPHONE (*Include Area Code*)<br>(1) Commercial<br>301-744-4700<br>(2) DSN<br>354-4700 |
|---|---|
| c. ADDRESS (*Include Zip Code*)<br>Standardization Team (Code 840M)<br>101 Strauss Avenue<br>Indian Head, MD 20640-5035 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Standardization Program Office, Attn: DLSC-LM<br>8725 John J. Kingman Road, Suite 2533<br>Fort Belvoir, VA 22060-6221<br>Telephone (703) 767-6888 DSN 427-6888 |

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